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(54) **EYEWEAR DEVICE LENS RETENTION MECHANISM**

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G02C 13/00	(2006.01)

(52) **U.S. Cl.**

CPC ***G02C 5/001*** (2013.01); ***G02C 11/10*** (2013.01); ***G02C 13/001*** (2013.01); *G02C 2200/06* (2013.01)

(58) **Field of Classification Search**

CPC G02C 2200/06; G02C 2200/08; G02C 2200/10; G02C 1/08

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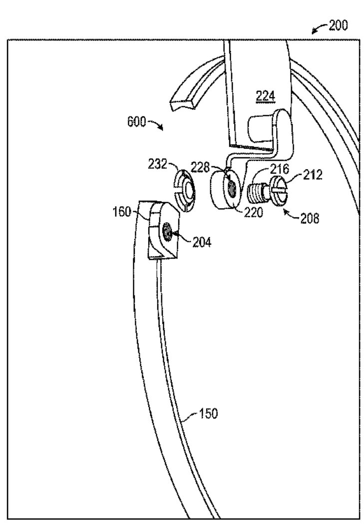
(57) **ABSTRACT**

An eyewear device has a lens retention mechanism that establishes electrical contact between onboard electronics and a lens ring by a resiliently compressible electrical contact member, such as a spring washer, located within a housing defined by a body of the eyewear device. The retention mechanism can further include a lock screw that is selectively engageable with the lens ring to allow disposal between retention and release of a lens by the lens ring, the lock ring being captured in the housing such as to resist removal of the lock screw from the housing when the lens ring is released for lens removal.

15 Claims, 9 Drawing Sheets



Related U.S. Application Data

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(58) **Field of Classification Search**
USPC 351/86, 92, 95–96, 106, 158; 349/13
See application file for complete search history.

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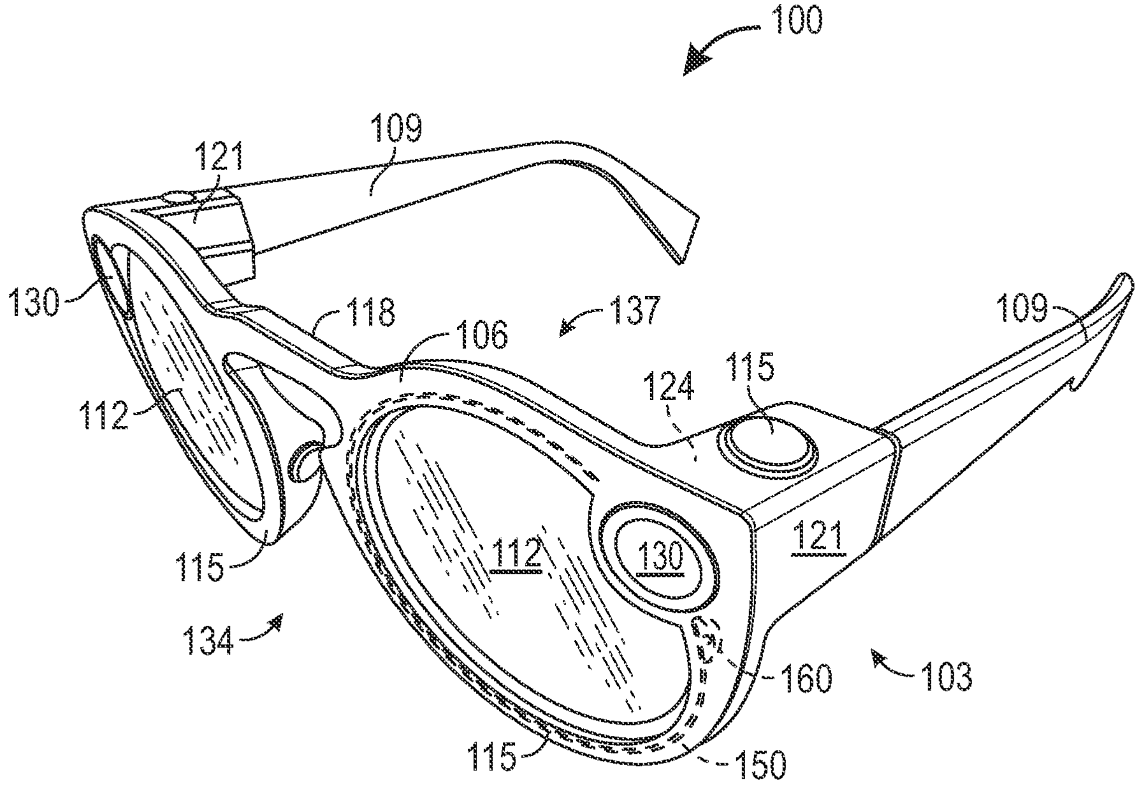
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FIG. 1

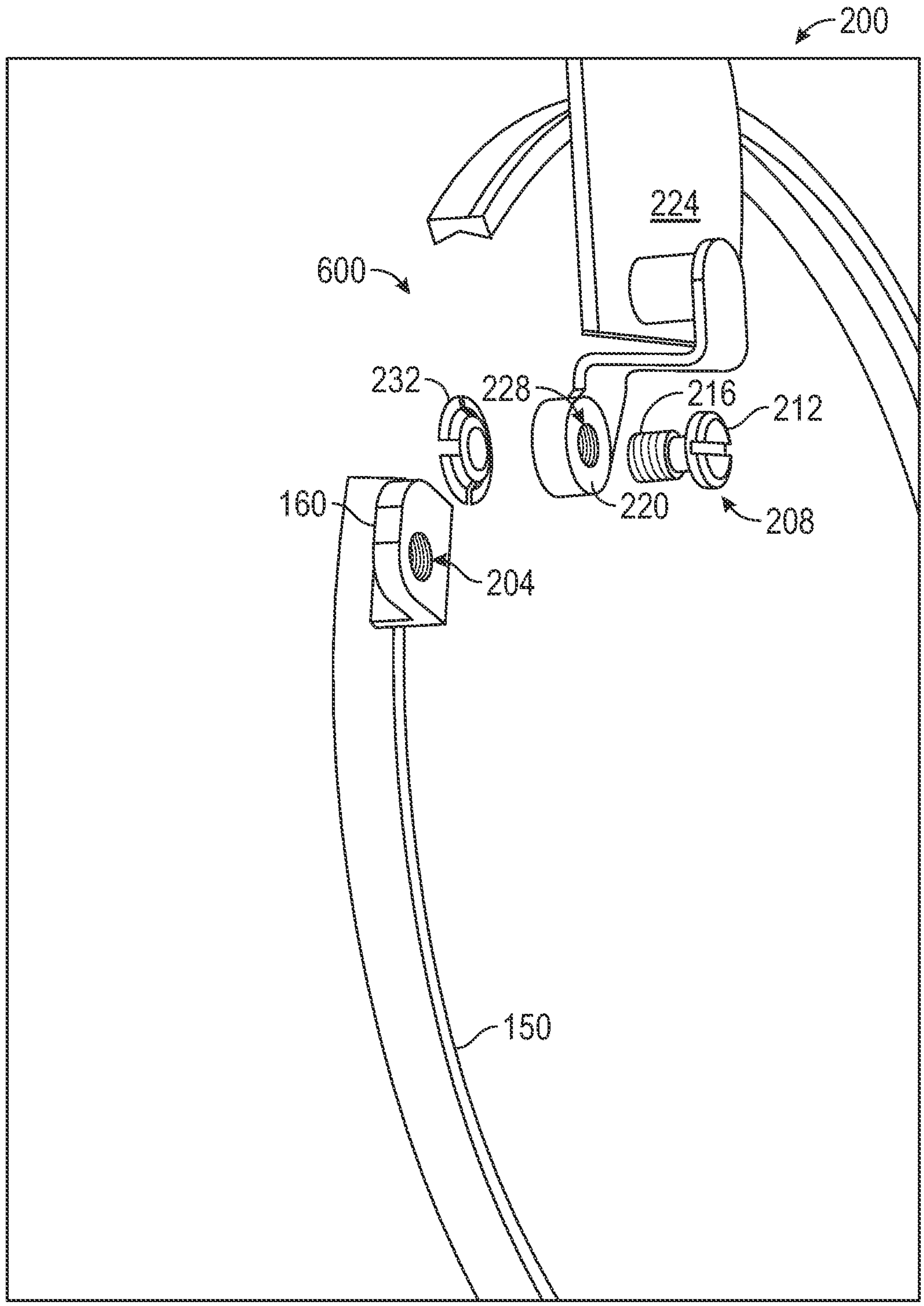

FIG. 2

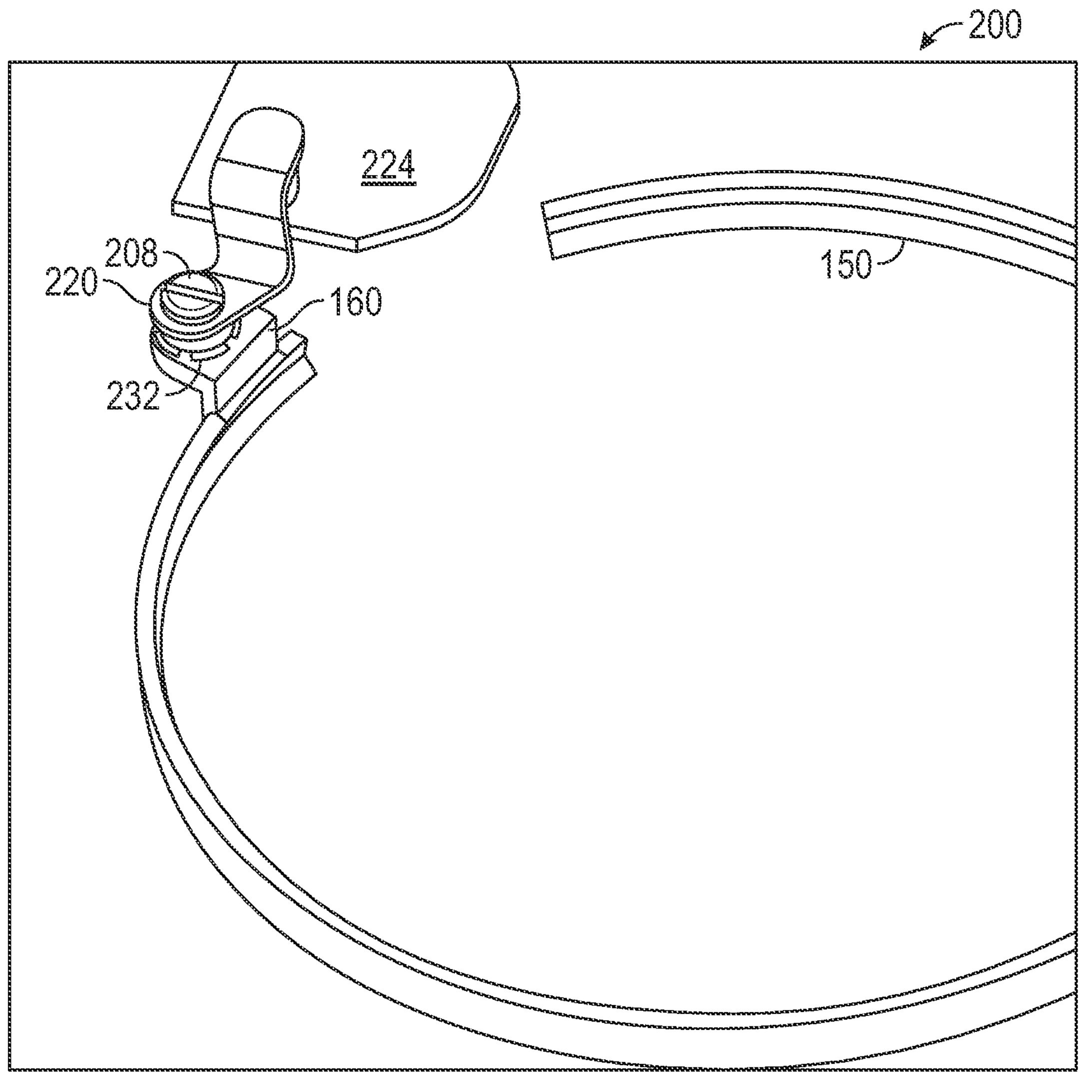

FIG.3

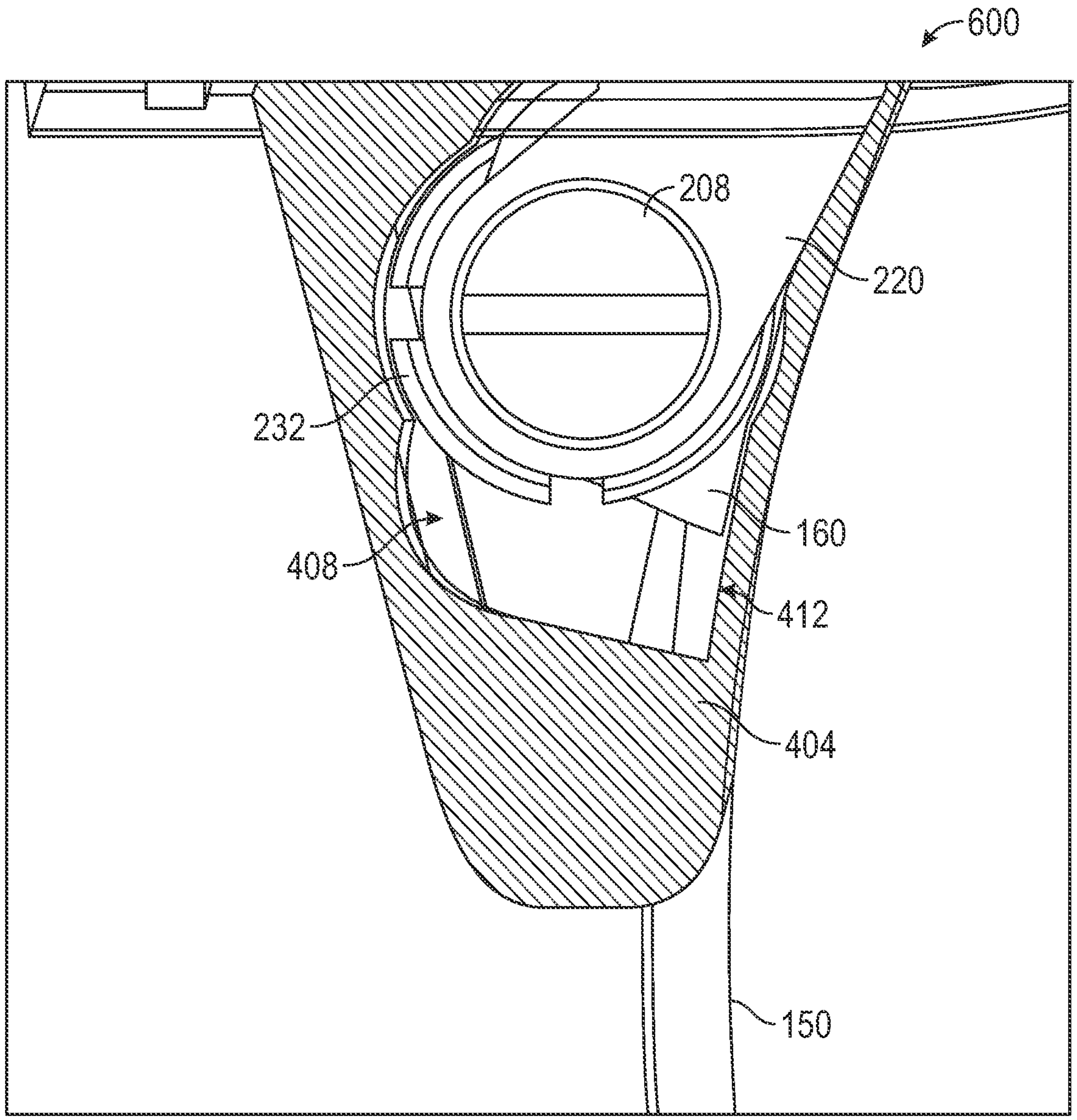

FIG. 4A

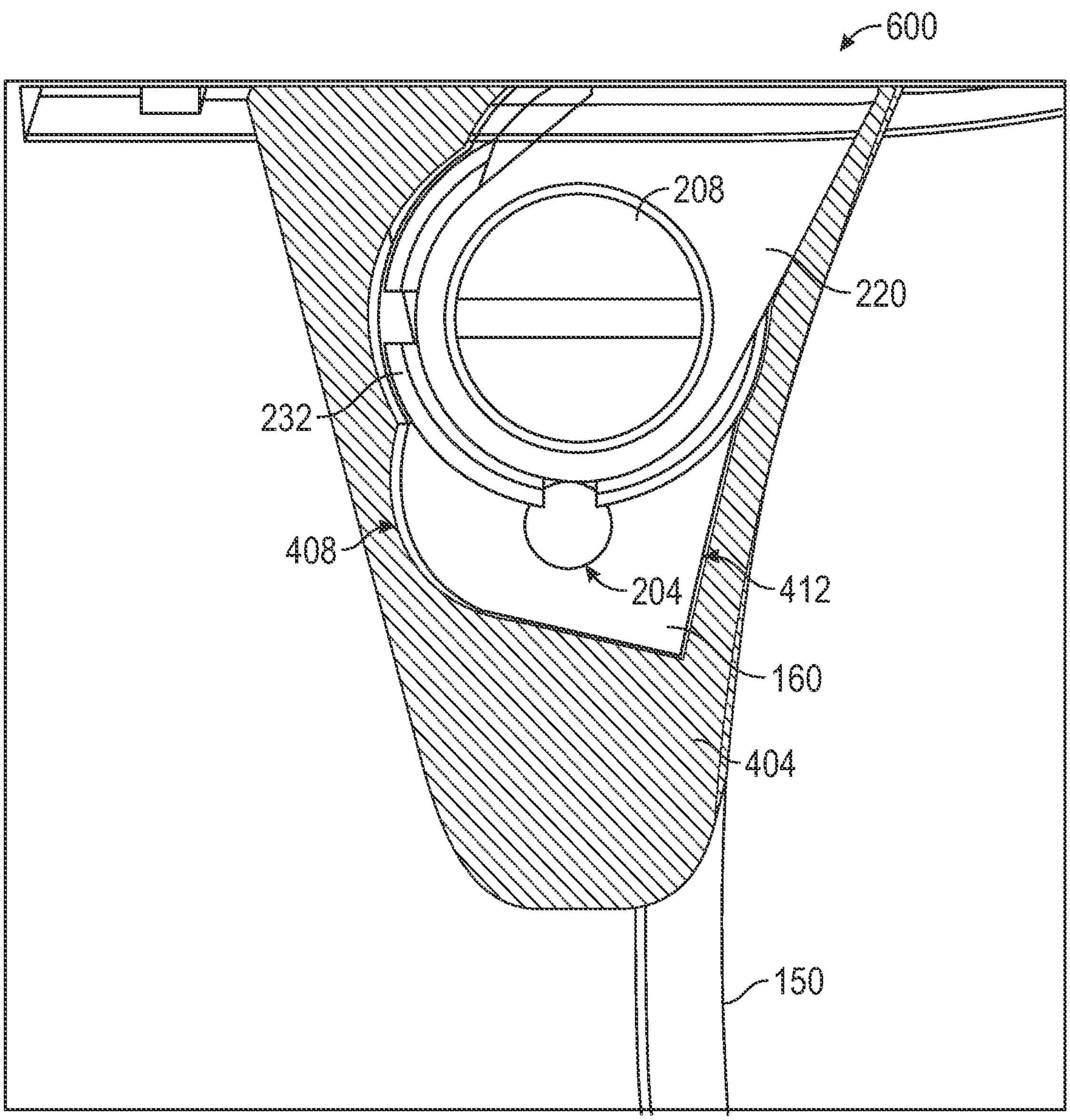

FIG. 4B

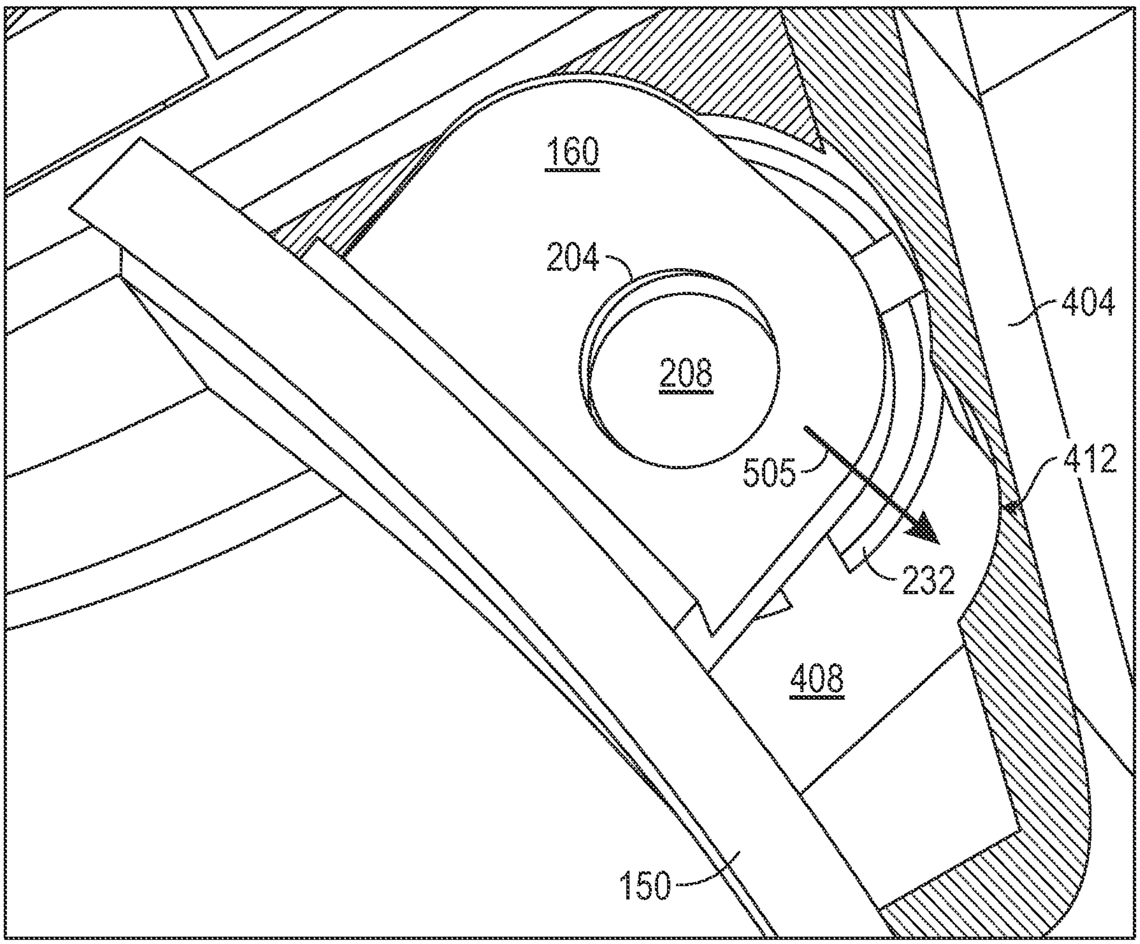

FIG. 5A

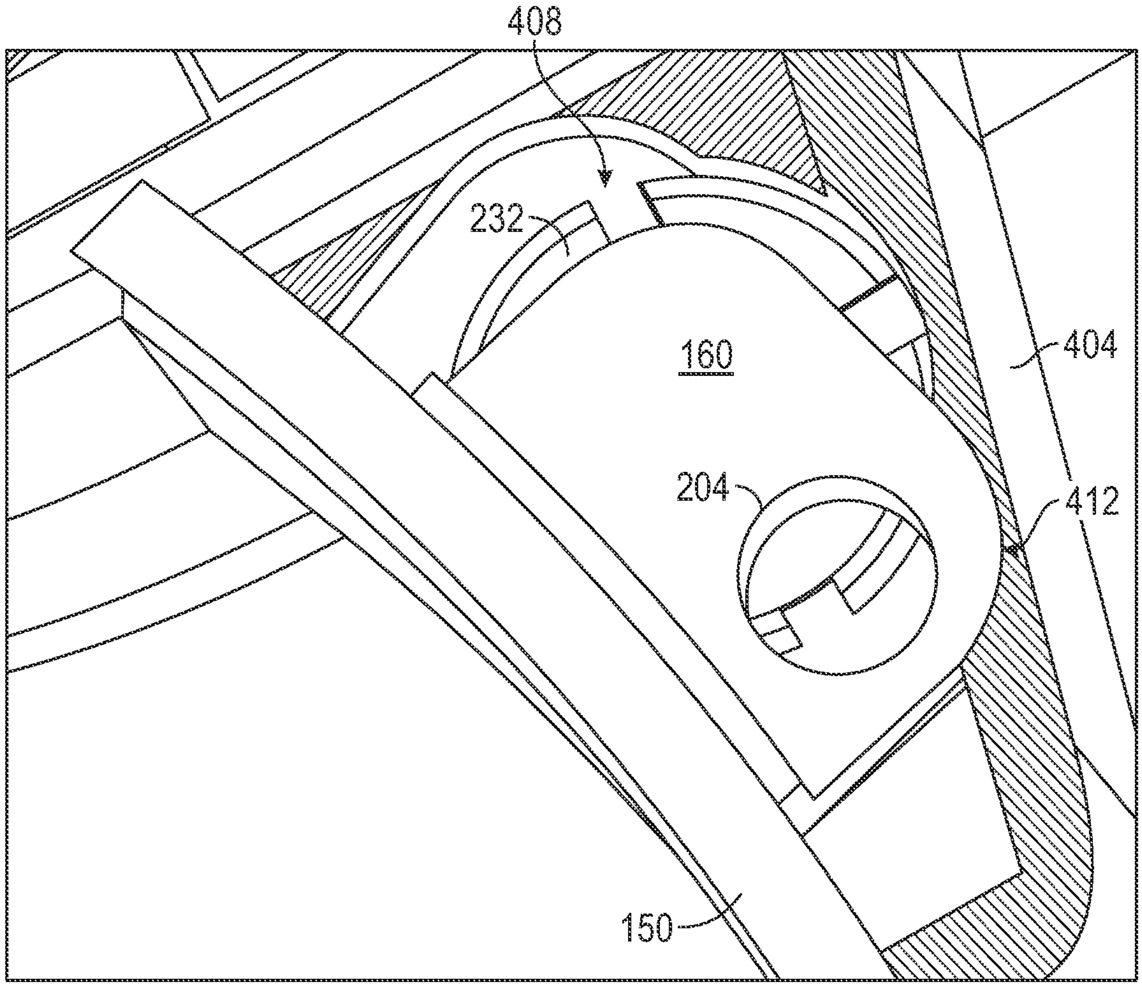

FIG. 5B

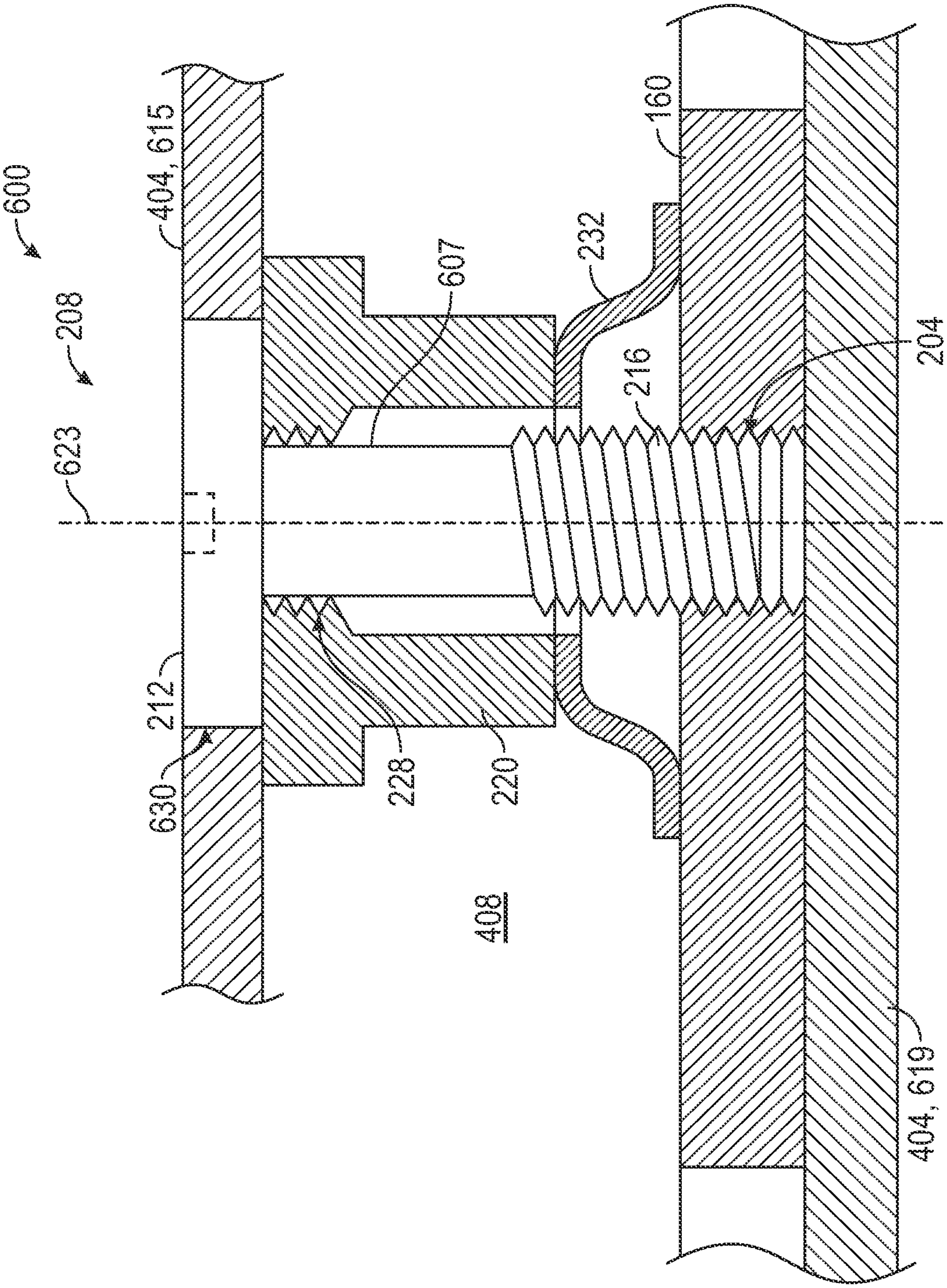
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FIG. 6A

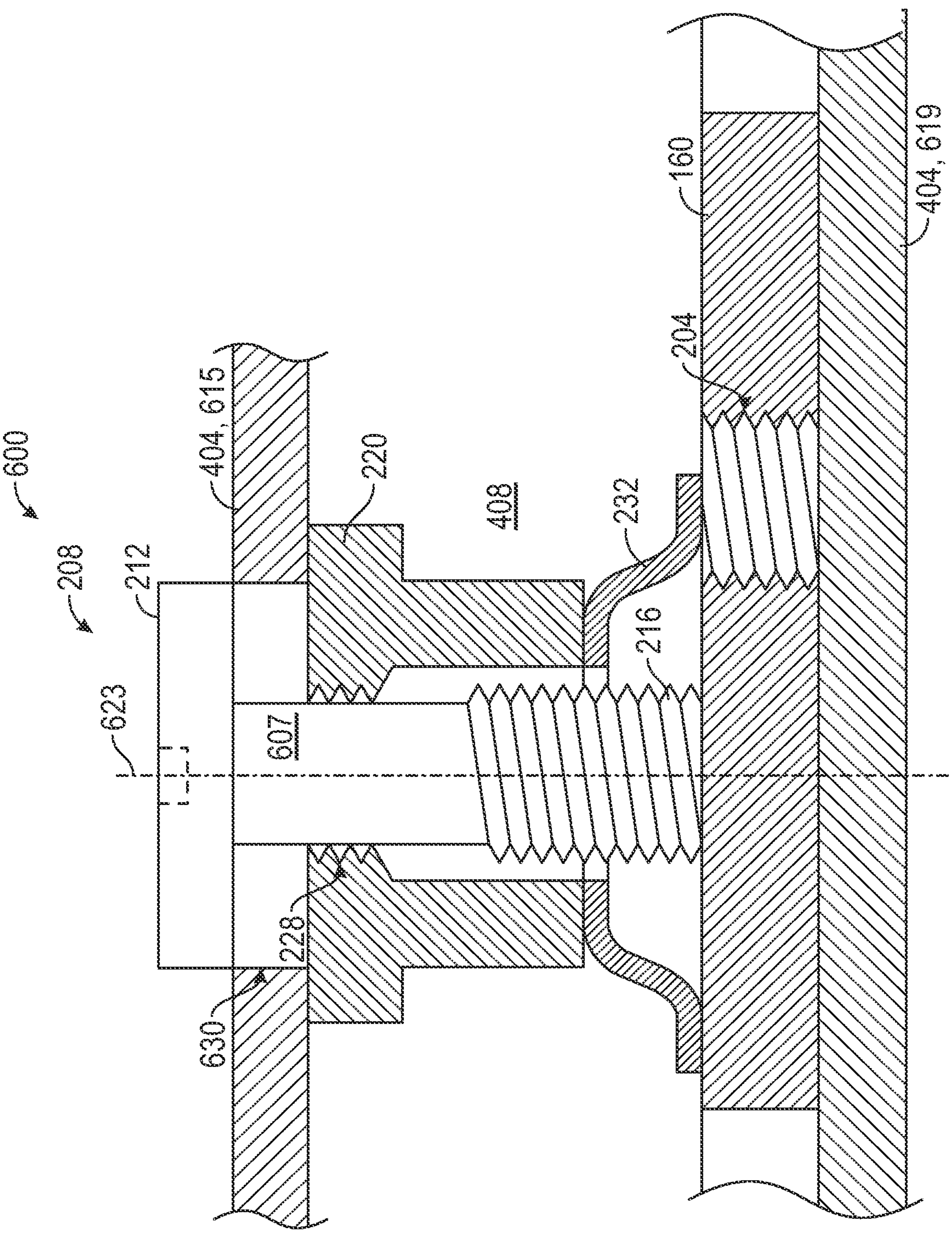
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FIG. 6B

EYEWEAR DEVICE LENS RETENTION MECHANISM

PRIORITY

This application is a continuation of U.S. application Ser. No. 16/256,681, filed Jan. 24, 2019, which application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 62/621,482, filed on Jan. 24, 2018, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

Electronics-enabled eyewear devices, such as smart glasses, have optical elements such as lenses held in view of the user by an eyewear frame. To allow for optional removal, repair, replacement, or changing of lenses, eyewear devices typically have retention mechanisms that are selectively lockable or unlockable by operation of a fastener such as a lock screw.

Such retention mechanisms often comprise a retainer element that extends peripherally around the lens, for example, comprising a retention ring. Some of the example embodiments disclosed herein provide for use of the retention element for electronics purposes, for example, relating to electromagnetic interference (EMI), electrostatic discharge (ESD), and/or signal transmission/reception considerations. Establishing an electronic connection between the retention element and onboard electronics of the eyewear device, while allowing for displacement of the retention element when the corresponding lens is to be removed or replaced is problematic owing to conflicting requirements for, on the one hand, selectively disengageable mechanical retention and, on the other hand, a reliable and consistent conductive connection.

BRIEF DESCRIPTION OF THE DRAWINGS

The appended drawings merely illustrate example embodiments of the present disclosure and cannot be considered as limiting its scope. To facilitate collation of numbered items in the description to the drawings, the first digit of each numbered item corresponds to the figure in which that item first appears. In the drawings:

FIG. **1** is a schematic of a three-dimensional view of an electronics-enabled eyewear device in the form of a pair of smart glasses, according to an example embodiment.

FIG. **2** is a three-dimensional exploded view of a lens retention assembly forming part of an eyewear device, according to one example embodiment.

FIG. **3** is a three-dimensional view of the lens retention assembly according to the example embodiment of FIG. **3**, the assembly for clarity of illustration being shown in an assembled form separate from a housing and lens rims forming part of the eyewear device.

FIG. **4**A is a schematic, partially sectioned rear view of a retainer locking mechanism forming part of a lens retention assembly, according to the example embodiment of FIG. **2**, the locking mechanism being located within a launch housing provided by an eyewear frame, and being shown in FIG. **4**A in a locked position.

FIG. **4**B is a view corresponding to FIG. **4**A, the retention locking mechanism being shown in an unlocked position.

FIG. **5**A is a partially sectioned front view of the retainer locking mechanism, according to an example embodiment of FIG. **4**A, the locking mechanism being shown in the locked position.

FIG. **5**B is a view corresponding to FIG. **5**A, the retainer locking mechanism being shown in an unlocked position.

FIG. **6**A is a schematic cross-sectional view of the retainer locking mechanism, according to an example embodiment of FIG. **2**, the locking mechanism being in a locked position.

FIG. **6**B is a view corresponding to FIG. **6**A, the locking mechanism being in an unlocked position.

The headings provided herein are merely for convenience and do not necessarily affect the scope or meaning of the terms used.

OVERVIEW

One aspect of the disclosure provides for a lens retention mechanism that establishes electrical contact between onboard electronics of an eyewear device and an elongated lens retainer (e.g., a retention ring) by a resiliently compressible electrical contact member (e.g., a spring washer) forming part of the lens retention mechanism.

In some embodiments, the lens retainer is conductively connected to the onboard electronics to serve as antenna element, the lens retention mechanism including an antenna launch that is conductively coupled to the onboard electronics and that is placed in electrical contact with the lens retainer by the electrical contact member, which is sandwiched in resilient compression between the lens retainer and the antenna launch. In some embodiments, the electrical contact is provided by a spring washer received on a lock screw that is screwingly engageable with a locking formation of the lens retainer.

The lens retention tension mechanism may further be configured and assembled within a housing defined by a body of the eyewear device such as to allow locking and unlocking of the lens retainer (e.g., to permit changing of lenses) while retaining the various components of the lens retention mechanism in a removal-resistant manner. In one such embodiment, a screw thread of the lock screw is located within the housing beyond a complementary screw-threaded opening, so that extraction of the lock screw from the housing is resisted by the screw-threaded opening. In an example embodiment, the screw-threaded opening that serves to capture the lock screw's screw-thread in the housing is defined by the antenna launch.

DETAILED DESCRIPTION

The description that follows includes devices, systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative embodiments of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the disclosed subject matter. It will be evident, however, to those skilled in the art, that embodiments of the disclosed subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

FIG. **1** shows an oblique front view of an electronics-enabled eyewear device **100** in the example form of a pair of smart glasses. The eyewear device **100** includes a body **103** comprising a front piece or frame **106** and a pair of temples **109** connected to the frame **106** for supporting the frame **106** in position on a user's face when the eyewear device **100** is worn. The frame **106** can be made from any suitable material such as plastics or metal, including any suitable shape memory alloy.

The eyewear device **100** has a pair of optical elements in the form of a pair of optical lenses **112** held by corresponding optical element holders in the form of a pair of lens rims **115** forming part of the frame **106**. The rims **115** are connected by a bridge **118**. In other embodiments, of one or both of the optical elements can be a display, a display assembly, or a lens and display combination. The eyewear device **100** can, in such embodiments, provide a virtual reality headset or an augmented reality display. Description in this example embodiment of elements relating to lens retention is thus to be read as, in other embodiments, being analogously applicable to different forms of optical elements that can be removably and replaceably received in the lens rims **115** by operation of a retention mechanism analogous to that described herein.

The frame **106** includes a pair of end pieces **121** defining lateral end portions of the frame **106**. In this example, a variety of electronics components are housed in one or both of the end pieces **121**, as discussed in more detail below. In some embodiments, the frame **106** is formed of a single piece of material, so as to have a unitary or monolithic construction.

The temples **109** are coupled to the respective end pieces **121**. In this example, the temples **109** are coupled to the frame **106** by respective hinges so as to be hingedly movable between a wearable mode (as shown in FIG. **1**) and a collapsed mode in which the temples **109** are pivoted towards the frame **106** to lie substantially flat against it. In other embodiments, the temples **109** can be coupled to the frame **106** by any suitable means. Each of the temples **109** includes a front portion that is coupled to the frame **106** and a suitable rear portion for coupling to the ear of the user, such as the curved earpiece illustrated in the example embodiment of FIG. **1**.

In this description, directional terms such as front, back, forwards, rearwards, outwards and inwards are to be understood with reference to a direction of view of a user when the eyewear device **100** is worn. Thus, the frame **106** has an outwardly directed front side **134** facing away from the user when worn, and an opposite inwardly directed rear side **137** side facing towards the user when the eyewear device **100** is worn. Similarly, the terms horizontal and vertical as used in this description with reference to different features of the eyewear device **100** are to be understood as corresponding to the orientation of the eyewear device **100** when it is level on the face of a user looking forwards. A horizontal or lateral direction of the eyewear device **100** thus extends more or less between the end pieces **121**, while a vertical or upright direction of the eyewear device **100** extends transversely to the horizontal direction, such that the lenses **112** have a more or less vertical or upright orientation.

The eyewear device **100** has onboard electronics **124** including a computing device, such as a computer, which can, in different embodiments, be of any suitable type so as to be carried by the body **103**. In some embodiments, various components comprising the onboard electronics **124** are at least partially housed in one or both of the temples **109**. In the present embodiment, various components of the onboard electronics **124** are housed in the lateral end pieces **121** of the frame **106**. The onboard electronics **124** includes one or more processors with memory, wireless communication circuitry, and a power source (this example embodiment being a rechargeable battery, e.g. a lithium-ion battery). The onboard electronics **124** comprises low-power, high-speed circuitry, and, in some embodiments, a display processor. Various embodiments may include these elements in different configurations or integrated together in different ways.

As mentioned, the onboard electronics **124** includes a rechargeable battery. In some embodiments, the battery is disposed in one of the temples **109**. In this example embodiment, however, the battery is housed in one of the end pieces **121**, being electrically coupled to the remainder of the onboard electronics **124**.

The eyewear device **100** is camera-enabled, in this example comprising a camera **130** mounted in one of the end pieces **121** and facing forwards so as to be aligned more or less with the direction of view of a wearer of the eyewear device **100**. The camera **130** is configured to capture digital still as well as digital video content. Operation of the camera **130** is controlled by a camera controller provided by the onboard electronics **124**, image data representative of images or video captured by the camera **130** being temporarily stored on a memory forming part of the onboard electronics **124**. In some embodiments, the eyewear device **100** can have a pair of cameras **130**, e.g. housed by the respective end pieces **121**. I

The eyewear device **100** further includes one or more input and output devices permitting communication with and control of the camera **130**. In particular, the eyewear device **100** includes one or more input mechanisms for enabling user control of one or more functions of the eyewear device **100**. In this embodiment, the input mechanism comprises a button **115** mounted on the frame **106** so as to be accessible on top of one of the end pieces **121** for pressing by the user.

The eyewear device **100** is, in this example embodiment, configured for wireless communication with external electronic components or devices, to which end the onboard electronics **124** is connected to an antenna integrated in the body **103** of the eyewear device **100**. In this example embodiment, the antenna is provided by a lens retainer in the example form of a lens ring **150** that additionally serves the purpose of removably and replaceably retaining the lens **112** in the corresponding lens rim **115**. Note that, in FIG. **1**, only one of the lens rims **115** is shown as having a corresponding lens ring **150** housed therein, but that both of the lens rims **115** is, in this example embodiment, provided with a respective lens ring **150** and associated lens retention mechanism, as described below.

In this example embodiment, the lens ring **150** is located in a circumferentially extending channel in a radially inner surface of the lens rim **115**, so that the lens ring **150** extends circumferentially around the majority of the periphery of the lens **112**, being engageable with the radially outer edge of the lens **112** to retain the lens **112** in the lens ring **150**. The lens ring **150** is disposable between a retention condition, in which it is tightened into contact with the radial edge of the lens **112** to keep it in the lens rim **115**, and a replacement condition in which the lens ring is somewhat dilated, to allow removal and replacement of the lens **112**. In this embodiment, the lens ring **150** has a locking formation in the form of a lock tab **160** to lock the lens ring **150** in the retention condition by engagement with a lock screw forming part of a retention locking mechanism, as will be described in greater detail below. Thus, loosening of the lock screw allows circumferential movement of the corresponding end of the lens ring **150**, to permit lens removal.

FIG. **2** shows an exploded or disassembled view of a lens retention assembly **200**, according to an example embodiment. The lens retention assembly **200**, in this example embodiment, comprises a lens retainer in the example form of the lens ring **150** (as previously described), and a locking mechanism **600** for selectively locking and unlocking the lens ring **150**, thereby disposing the lens ring **150** to the retention condition or the replacement condition, as the case may be. Operation of the locking mechanism **600** is described in greatest detail later below with reference to FIGS. **6**A and **6**B. As will emerge from the description that follows, the locking mechanism **600**, in this example embodiment, serves the dual purposes of (a) selective locking/unlocking of the lens ring **150**, and (b) establishing a reliable electrical connection between the lens ring **150** and the onboard electronics **124** of the eyewear device **100**, to enable use of the lens ring **150** as an antenna element for sending and receiving electromagnetic signals.

Note that, to allow a clearer view of the various components of the lens retention assembly **200**, a launch housing **404** (see FIG. **4**) that contains a number of the components of the locking mechanism **600** is omitted in FIGS. **2** and **3**. As will be seen more clearly with respect to the descriptions of FIGS. **4**-**6**, the housing **404**, is in this example embodiment provided by the body **103** of the eyewear device **100**, forming part of a respective one of the end pieces **121**.

The locking mechanism **600**, in this example embodiment, achieves locking of the lens ring **150** by engagement of a fastener in the form of a lock screw **208** with the lens ring **150**. In particular, a screw thread **216** of the lock screw **208** is screwingly receivable in a complementary screw-threaded lock hole **204** defined in the lock tab **160** at a free end of the lens ring **150**. As can be seen in FIG. **2**, the orientation of the lock screw **208** and the lock hole **204** is, in this embodiment, such that a longitudinal axis (indicated in FIG. **6**A as item **623**) of the lock screw **208** is oriented in the fore-and-aft direction of the eyewear device **100**. Turning briefly to FIG. **6**A, it will be seen that screwing engagement of the lock screw **208** with the lock tab **160** serves to anchor the lock tab **160** relative to the frame **106** by location of a screw head **212** in a close-fitting complementary socket **630** in a rear wall **615** of the housing **404**. In this position, the lock screw **208** and the lock tab **160** of the lens ring **150** are anchored against transverse movement relative to the housing **404** by engagement of the screw head **212** with the housing **404**. In this description, unless otherwise indicated, the term axial refers to a direction substantially parallel to the longitudinal axis **623** of the lock screw **208**. Similarly, the term transverse, in these contexts, refers to directions that are transverse to the screw axis **623**.

Returning now to FIG. **2**, it will be seen that the locking mechanism **600** further includes an electronics launch in the example form of an antenna launch **220** that it is conductively coupled to a printed circuit board (PCB) **224** forming part of the onboard electronics **124** of the eyewear device **100**. The antenna launch **220** is a metal component of a material that can be selected for its electrical conductivity properties, in this example embodiment being a copper component. The antenna launch **220** is thus at one end thereof electrically connected to the PCB **224**, at its other end defining a screw-threaded passage co-axial with and complementary to the screw thread **216** of the lock screw **208**. During assembly, the screw thread **216** is screwed entirely through the passage **228** of the antenna launch **220** (see, for example, FIG. **6**A), so that a smooth shank **607** of the screw head **212** is located co-axially in the screw-threaded passage **228**, lock screw's screw thread **216** and screw head **212** being located axially on opposite sides of the passage **228**. As will be described later, the screw-threaded passage **228** of the antenna launch **220** serves to hold the screw thread **216** captive in the housing **404** by resisting axial extraction of the lock screw **208** from the housing **404**.

The lens retention assembly **200** further includes a resiliently compressible contact element in the form of a cup-shaped spring washer **232** co-axial with the lock screw **208** and axially sandwiched between the antenna launch **220** and the lock tab **160** of the lens ring **150**. As can best be seen in FIG. **6**A, the lock screw **208** extends axially through a central opening of the spring washer **232**, thus holding the spring washer **232** captive against transverse escape. The spring washer **232** is in direct physical contact with both the lock tab **160** (bearing against a flat surface provided by the lock tab **160**) and the antenna launch **220**, thus providing a conductive path between the antenna launch **220** and the lock tab **160**. It will be appreciated that the spring washer **232** in this manner electrically couples the lens ring **150** (via the lock tab **160**) to the PCB **224** (via the antenna launch **220**).

Moreover, in this example embodiment, electric coupling of the lens ring **150** to the PCB **224** via the spring washer **232** is promoted by selecting a relatively highly conductive metal for the spring washer **232**. In this example embodiment, the spring washer **232** is of copper. It will be appreciated that the lock screw **208**, being in physical contact with both the antenna launch **220** and the lock tab **160** also, to some extent, provides an electrical connection between these components. However, the lock screw **208** in this example embodiment has greater resistivity than the spring washer **232**, being of mild steel as opposed to the copper pathway provided by the spring washer **232**. Note that the lens retention assembly **200** is designed and configured such that the electrical connection between the PCB **224** and the lens ring **150** is fully operational purely via the spring washer **232**, so that electrical coupling of the lens ring **150** to the PCB **224** is not dependent on the lock screw **208**.

Consistent and effective conductive coupling of the spring washer **232** to both the antenna launch **220** and the lock tab **160** of the lens ring **150** is promoted in the example embodiment of FIG. **6**A by virtue of the spring washer **232** by default being in a resiliently compressed state. Note that, in this example embodiment, the front wall **619** and the rear wall **615** of the launch housing **404** do not move relative to each other, so that an axial spacing between the rear wall **615** and the front wall **619** is substantially consistent. The axial length of the antenna launch **220** and the thickness of the lock tab **160** are likewise consistent, defining a substantially consistent axial spacing between them, in which the spring washer **232** is located. An unstressed axial height of the spring washer **232** is somewhat greater than the available space between the antenna launch **220** and the lock tab **160**, so that the spring washer **232** is, in operation, axially compressed both in the locked position (FIG. **6**A) and in the unlocked position (FIG. **6**B). Due to his axial compression, the spring washer **232** resiliently biases the antenna launch **220** and the lock tab **160** in opposite axial directions, urging them apart. It will be appreciated that such enforced physical engagement of the spring washer **232** against the antenna launch **220** and the lock tab **160** ensures the provision of a consistent and uninterrupted conductive path.

Turning briefly to FIG. **3**, therein is shown the lens retention assembly **200** in fully assembled form and in the retention condition in which lock screw **208** is in the locked position in which its screw thread **216** is received in the screw-threaded lock hole **204** of the lock tab **160**. It will be noted (as can also be seen in FIGS. **6**A and **6**B) that the screw head **212** bears directly on the antenna launch **220**, so that the additional tightening of the lock screw **208** can further compress the spring washer **232** by axially inward movement of the antenna launch **220**. Note, however, that the length of the lock screw **208** and the height of the screw head **212** are selected such that movement of the screw head **212** fully through the rear wall **615** is not possible.

Turning now to FIGS. **4**A and **4**B, therein is shown a rear view of the locking mechanism **600** and launch housing **404** (i.e., seen from the rear side **137** of the eyewear device), with the rear wall **615** of the housing **404** being shown as removed to reveal the interior of the housing **404** and the components therein. The lock tab **160** is shown in FIG. **4**A in the locked position and in FIG. **4**B in the unlocked position.

The interior of the housing **404** defines a shaped housing cavity **408** that includes a channel **412** in which the lock tab **160** is slidable to move between the locked position (FIG. **4**A) and the unlocked position (FIG. **4**B). The shaped housing cavity **408** also defines a part circular space in which the spring washer **232** is held captive against transverse movement, preventing transverse sliding of the spring washer **232** into the sliding channel **412** for the lock tab **160**. Note that the lens ring **150** is resiliently contractible, being biased, spring-fashion, to the position shown in FIG. **4**B, in which the lock tab **160** stops against a side wall of the housing cavity **408**.

FIG. **5**A shows a front view of the locking mechanism **600** and the housing **404**, with the front wall **619** being omitted for clarity of illustration. As can be seen in FIG. **5**A, the lock tab **160** (when it is in the locked position) is held against transverse movement in the direction of arrow **505** my location of the lock screw **208** in the lock hole **204**. When the lock screw **208** is screwed out of engagement with the lock hole **204** (see also FIG. **6**B), the lens ring **150** resiliently expands, moving the lock tab **160** into its unlocked position shown in FIG. **5**B. In this expanded position, the diameter of the lens ring **150** is somewhat greater than in the locked position (FIG. **5**A), allowing removal and replacement of the lens **112**. Note that when the lock tab **160** slides transversely into the unlocked position, the spring washer **232** remains in its original position (and does so even in the absence of the lock screw **208**) due to the shaped periphery of the housing cavity **408**.

In use, initial assembly of the locking mechanism **600** includes locating the spring washer **232**, antenna launch **220**, and lock tab **160** in the housing cavity **408**, and thereafter screwing the screw thread **216** of the lock screw **208** through the screw-threaded passage **228** of the antenna launch **220**. The lens ring **150** is then manually compressed or contracted, bringing the lock hole **204** of the lock tab **160** into register with the lock screw **208**. Screwing the lock screw **208** into the lock hole **204** then places the locking mechanism **600** into the locked position, keeping the lens ring **150** in its retention condition (FIGS. **4**A, **5**A, and **6**A).

When the lens **112** is to be removed the screw **208** is unscrewed, extracting its screw thread **216** axially from the lock hole **204**. Responsive to the unscrewing, the lens ring **150** automatically expands into the unlocked position (FIGS. **4**B, **5**B, and **6**B).

Note that in both the locked and unlocked position, the antenna launch **220** is electrically coupled to the lens ring **150** via the spring washer **232**, allowing communication of electrical signals between the PCB **224** and the lens ring **150** for transmission and reception of electromagnetic signals by the lens ring **150** as an antenna.

A benefit of the example locking mechanism **600** is that all of its components are held captive within the housing **404**, being resistant to user removal and/or tampering. In particular, simple axial extraction of the lock screw **208** is frustrated by fouling of its screw thread **216** against the screw thread of the passage **228** through the antenna launch **220**. Thus, the lock screw **208** can be removed from the housing **404** only by screwing its screw thread **216** through the passage **228** of the antenna launch **220**, which requires screwing engagement between the screw thread **216** and the passage **228**. Such engagement, however, is frustrated by the difficulty of applying an axial load to the lock screw **208** in order to effect such screwing engagement. It will be remembered that the lock screw **208** is relatively small (considering, for example the scale of the eyewear device **100** as shown in FIG. **1**), so that its weight is typically insufficient to reliably effect screwing engagement of the screw thread **216** with the passage **228**.

In this example embodiment, the screw head **212** is substantially flush with the outer surface of the rear wall **615** of the housing **404** in the locked position (FIG. **6**A), a substantially flush with the exterior surface of the rear wall **615**, but stands proud of the rear wall **615** in the unlocked position (FIG. **6**B). Due to the relatively small size of the lock screw **208** and the limited height of the screw head **212**, gripping of the screw head **212** with a tool such as pair of pliers in order to pull the lock screw **208** axially outwards with sufficient force to cause initiation of screwing engagement with the antenna launch **220** is difficult, if not unfeasible. In other example embodiments, the dimensions of the relevant components are chosen such that the screw head **212** is flush or sub flush with the rear wall **615** in the unlocked position. In some such embodiments, the length of the screw thread **216** is selected such that, at an extreme extracted position of the lock screw **208** (at which position the screw thread **216** fouls on the screw thread of the launch passage **228**), the screw head **212** is, in some embodiments, flush or sub flush or, in other embodiments, not fully displaced out of socket **630** defined therefor in the rear wall **615**.

As will be evident from the description of the example embodiment above, this disclosure provides for a lens ring assembly that achieves electrical contact by use of a spring washer **232**, allowing the lens ring **150** to be unscrewed and extended to permit the replacement of lenses **112** while retaining the relevant components (e.g., the lens ring **150**, the lock screw **208**, and the antenna launch **220**) in a tamper-resistant manner that prevents or resists tampering, damage, or loss of individual components.

This is in contrast to existing lens retention assemblies, in which unscrewing of the lock screw **208** can be accompanied by intentional or accidental removal of the lock screw **208**. Loss or misplacement of the lock screw **208** after such extraction is a common occurrence that leads to loss of functionality of the assembly.

A further benefit of the described example embodiments is that the electrical contact provided by the spring washer **232** is superior in quality and reliability to alternative mechanisms, such as provision of an electrical path provided by the lock screw **208**. In such a mechanism, the quality of the connection is closely dependent on the particular torque or tension at which the lock screw **208** is fastened, which can require sophisticated and expensive tools for proper assembly and operation. In contrast, the electrically conductive path provided by the spring washer **232** is tolerant of (and may indeed be agnostic to) variations in the screw tension or applied torque.

Language

Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

Although an overview of the disclosed matter has been described with reference to specific example embodiments, various modifications and changes may be made to these embodiments without departing from the broader scope of embodiments of the present disclosure. Such embodiments of the inventive subject matter may be referred to herein, individually or collectively, by the term "invention" merely for convenience and without intending to voluntarily limit the scope of this application to any single disclosure or inventive concept if more than one is, in fact, disclosed.

The embodiments illustrated herein are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed. Other embodiments may be used and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. The Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

As used herein, the term "or" may be construed in either an inclusive or exclusive sense. Moreover, plural instances may be provided for resources, operations, or structures described herein as a single instance. Additionally, boundaries between various resources, operations, modules, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in a context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within a scope of various embodiments of the present disclosure. In general, structures and functionality presented as separate resources in the example configurations may be implemented as a combined structure or resource. Similarly, structures and functionality presented as a single resource may be implemented as separate resources. These and other variations, modifications, additions, and improvements fall within a scope of embodiments of the present disclosure as represented by the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

What is claimed is:

1. An eyewear device comprising:
- an eyewear frame;
- onboard electronics housed by the eyewear frame;
- an eyewear lens mounted on the eyewear frame;
- a lens retainer by which the eyewear lens is releasably retained on the eyewear frame, the lens retainer comprising a lens ring that extends circumferentially along a periphery of the eyewear lens, the lens ring being an electrically conductive component that is configured for co-operative connection to the onboard electronics and that is disposable between:
  - a retention condition in which the lens ring engages a peripheral edge of the eyewear lens to prevent removal of the eyewear lens from the eyewear frame, and in which a releasable end of the lens ring is anchored against circumferential movement relative to the eyewear lens to effect dilation of the lens ring, and
  - a replacement condition in which the lens ring is dilated relative to the retention condition, permitting removal and/or replacement of the eyewear lens, disposal of the lens ring from the retention condition to the replacement condition comprising circumferential movement of the releasable end of the lens ring relative to the eyewear lens to cause dilation of the lens ring; and
- a locking mechanism that provides a releasable physical coupling and a persistent electrical coupling with the lens ring, the locking mechanism comprising:
  - a fastener configured for engagement with a complementary locking formation defined by the lens ring, the fastener being operable between:
    - a locked condition in which the fastener is engaged with the locking formation of the lens ring such that the fastener anchors the releasable end of the lens ring circumferentially in a position in which the lens ring is in the retention condition, thereby releasably locking the lens ring in the retention condition, and
    - an unlocked condition in which the fastener is free from the locking formation of the lens ring, permitting circumferential movement of the releasable end of the lens ring relative to the eyewear lens to allow disposal of the lens ring from the retention condition to the replacement condition; and
  - a conductive contact element that is electrically connected to the onboard electronics and that is in engagement with the lens ring to complete a conductive path which provides an electrical connection between the onboard electronics and the lens ring; and
    - a spring-loaded bias mechanism configured to resiliently urge the contact element towards contact engagement with the lens ring, thereby to maintain said electrical connection both when the lens ring is locked in the retention condition and when the lens ring is unlocked and is in the replacement condition.

2. The eyewear device of claim **1**, wherein the fastener defines an axis along which the fastener is axially displaceable to dispose the fastener between the locked condition and the unlocked condition, wherein the contact element is resiliently compressible in the axial direction of the fastener, the contact element being axially sandwiched between, on the one hand, an electronics launch of the onboard electronics and, on the other hand, the locking formation of the lens ring, so that the contact element conductively bridges an axial spacing between the electronics launch and the locking formation of the lens ring, resilient action of the spring-loaded bias mechanism being caused by resilient resistance of the contact element to axial compression.

3. The eyewear device of claim **2**, wherein the contact element defines an opening through which the fastener extends axially, such that the contact element is held captive by the fastener against transverse movement in the axial spacing.

4. The eyewear device of claim **3**, wherein the contact element is a spring washer.

5. The eyewear device of claim 4, further comprising a capture mechanism configured to resist axial extraction of the fastener from the eyewear device when the fastener is in the unlocked condition.

6. The eyewear device of claim 5, wherein the fastener is a screw having a screw thread that, in the locked condition, is received in a complementary screw-threaded hole defined by the lens ring locking formation.

7. The eyewear device of claim 6, wherein the electronics launch is axially captured between the spring washer and a head of the screw that is accessible from an exterior of the eyewear frame, so that tightening of the screwing engagement with the lens ring locking formation urges the electronics launch towards the lens ring locking formation.

8. The eyewear device of claim 7, wherein the electronics launch has a screw-threaded opening complementary to the screw thread, the screw head and the screw thread of the screw being located on opposite axial sides of the electronics launch, so that axial extraction of the screw is resisted by the screw-threaded opening of the electronics launch.

9. The eyewear device of claim 6, wherein, in the unlocked condition, the locking formation is transversely slidable relative to the screw and the spring washer held captive thereby, allowing circumferential movement of the releasable end of the lens ring to dispose the lens ring to the replacement condition.

10. The eyewear device of claim 6, wherein a screw head of the screw is located in a complementary recess defined by an electronics launch housing forming part of the eyewear locking formation, such that an axially outer surface of the screw is sub-flush relative to an exterior surface of the electronics launch housing.

11. An eyewear device comprising:

an eyewear frame that comprises a lens holder configured to hold a lens;

a processor device housed by the eyewear frame and configured to provide one or more electronics-enabled functionalities to the eyewear device;

a metal retention ring housed by the lens holder and disposable between a retention condition in which it mechanically retains the lens in the lens holder, and a replacement condition in which removal or replacement of the lens is permitted;

a locking formation fixed to the retention ring;

an electrically conductive electronics launch element that is conductively coupled to the processor device;

an elongate fastener that passes axially through the electronics launch element and that is engageable with the locking formation to lock the retention ring in the retention condition; and

a spring washer that is received on the fastener and that is sandwiched between the retention ring locking formation and the electronics launch element, the spring washer being in a resiliently compressed state in which it bears against both the electronics launch element and the retention ring locking formation, thereby providing an electrically conductive connection between retention ring and the electronics launch element.

12. The eyewear device of claim 11, wherein the fastener is a lock screw that is screwingly engageable with the locking formation to lock the retention ring in the retention condition.

13. The eyewear device of claim 12, wherein at least part of the lock screw is held captive such as to resist axial extraction of the lock screw from the electronics launch element.

14. The eyewear device of claim 13, wherein the lock screw is held captive against axial extraction by a screw thread of a complementary screw-threaded opening through which the lock screw extends co-axially such that a screw thread of the screw and a head of the lock screw are located on axially opposite sides of the screw-threaded opening.

15. The eyewear device of claim 14, wherein the screw-threaded opening that holds captive the screw-threaded of the lock screw is provided by the electronics launch element.

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